

THE OPPORTUNITY

Japan has every ingredient needed to lead the global energy transition: world-class research institutions, patient corporate capital, engineering and manufacturing excellence across the industries most central to decarbonisation, a stable government committed ¥150 trillion through 2050, and an industrial base ready to absorb and scale new technology.

And yet Japan is falling behind. Climate-tech startup formation lags the US and Europe by a wide margin. Entrepreneurs remain few, and those with multi-company scaling experience fewer still. Foreign innovators with proven technology rarely prioritise Japan — the market is too difficult to enter without dedicated support.

The missing piece is deployment infrastructure: the permitting pathways, community consent frameworks, and blended-finance structures that enable proven technology to reach the ground at scale. Without it, no amount of capital or R&D resolves the stall between lab and market.

WHAT ARCS DOES

ARCS (Accelerated Readiness for Climate Solutions) is a **First-of-a-Kind Deployment Organisation**. It builds and operates the shared infrastructure that no single company, utility, or government agency can create alone — a reusable operating system for getting climate technology deployed in Japan.

It is not a funder, an accelerator, or a policy body. It is the connective layer between them.

The Kaizen Venture Model: rather than sending Japanese founders abroad to Silicon Valley accelerators, ARCS brings the world's best climate founders to Japan. Running their technology through Japan's industrial base, R&D depth, and manufacturing rigour produces a Japan-hardened technology: more refined, more deployable, globally competitive. Local people hired in the process become Japan's next generation of climate entrepreneurs.

Mission: make Japan a hub for climate-technology deployment and scaling, while building the next generation of domestic founders.

Method: Global Founder × Japanese Industrial Partner × Deployment OS

CASE STUDY: GEOTHERMAL AS THE ANCHOR THEME

Geothermal illustrates both Japan's structural advantage and its deployment failure in the sharpest possible terms. Japan holds the world's third-largest geothermal resource, and its equipment manufacturers — MHI, Toshiba, and Fuji Electric — supply 65–75% of global turbines. Despite this, less than 3% of that potential is online. The bottleneck is the lack of infrastructure that allows for risk-taking behaviour — not access to technology.

The unlock is Closed-Loop Geothermal Systems (CLGS): fluid circulates in a sealed pipe with no groundwater extraction and near-zero seismicity. This removes the scientific and legal basis that onsen operators have used to block development under the Hot Springs Act of 1948 for decades. Eavor Technologies demonstrated commercial grid delivery in Germany in December 2024. Japanese corporates are already engaged: Chubu Electric and Mizuho are Eavor investors; Mitsui and Chevron are piloting CLGS in Hokkaido; METI's Next-Gen Geothermal Council has 70+ enrolled companies. The technology is ready and needs more site to scale. What is missing is the deployment OS to put it in the ground — the same OS that, once built on geothermal, transfers to every sector ARCS takes on next.

THE ASK

\$10–15M philanthropic · 3 years

Structured to catalyse 3+ co-funding from NEDO, JOGMEC, and corporate PPA partners — targeting more than \$45M in total capital deployed by Year 5.

PROPOSER

Tomo Shibata — Former Founding Partner, Japan Energy Capital. Built and ran 15 deep-tech climate investments: Eavor, EnergyDome, Svante, GHGSat. Co-led GSAP with JETRO, New Energy Nexus, Third Derivative. Prior: Manager, Bain & Company Zurich. Cambridge (Physical Geography) · UCL (Environmental Economics). Languages spoken: EN · JP · FR.

THE FOUR BARRIERS

A technology that clears TRL 4 immediately runs into the same structural wall in Japan, regardless of sector. In a culture where risk-taking is rarely rewarded, the incentive to push through without dedicated support simply is not there.

Barrier	Description	Evidence
01 Capital Gap	Capital exists but is fragmented. Non-dilutive grants are growing yet remain far smaller than in the US, and navigating the public funding landscape without a guide is a significant burden on its own.	NEDO grant magnitude is a fraction of US DOE equivalents. No risk-sharing instrument exists between R&D grants and private growth equity for ¥1–10B FOAK pilots.
02 No Host Sites	No entity is mandated to carry the operational risk of hosting a first-of-a-kind demonstration. Sites require land, grid interconnect, water rights, and regulatory cover — simultaneously.	For example Japan holds 23 GW of geothermal potential. In nine years, only one new binary plant has come online.
03 Permitting Maze	No accelerated multi-agency pathway exists for novel technology. 47 prefectures each interpret national regulation differently, and no single agency is mandated to resolve the grey zones that result.	Offshore wind: 6–8 years to permit in Japan vs. the EU's 2-year statutory cap.
04 No Playbook	No institution captures what each deployment teaches. Permit timelines, community consent dynamics, finance structures — every team rebuilds from zero when it starts.	After 30+ years of Japanese geothermal activity, there is still no shared community engagement template across the industry.

These barriers are not technology-specific. They recur across many sectors such as geothermal, offshore wind, methane monitoring, industrial heat, long-duration storage, and carbon capture. They are a systemic problem requiring a systemic solution.

THE DEPLOYMENT OS: FOUR MODULES, BUILT ONCE

ARCS builds four reusable infrastructure modules — the **DEPLOYMENT OS** — each designed to neutralise one of the four barriers above. The modules are technology-agnostic and the core infrastructure is shared across sectors. The hardest build happens once, on the anchor theme; every sector that follows inherits it at near-zero marginal cost.

Module	What It Does	Y1 Metric	3-Year Target
Permit Router	Builds a shared multi-agency routing engine. Eliminates the per-project overhead of litigating permit interpretations from scratch in each of 47 prefectures.	Sub-12-month pathway vs. 6–10yr baseline	Shared decision tree adopted across all ARCS partners
Community Playbook	Converts consent-building experience into replicable templates that transfer across every subsequent site and sector. Each engagement updates the Playbook for the next.	≥2 Onsen Kumiai (for Geothermal) under formal Playbook protocol	Playbook v2; 3+ completed engagements
Pilot Finance Stack	Pre-assembles blended-finance toolkit combining JOGMEC guarantees, NEDO grants, utility PPAs, and philanthropic first-loss. Bridges the gap between R&D grants and private equity.	Finance package closed before first drill date	3×+ NEDO/JOGMEC leverage per philanthropic \$
Learning Loop	Converts each completed pilot into measurable improvements to the other three modules, making the OS progressively cheaper and faster with every deployment.	OS cost ≥40% below Y1 by Year 3	METI/JOGMEC formal OS adoption; public dashboard

WHO ARCS SERVES

Stakeholder	Value Delivered
Japanese climate startup ecosystem	A deployment pathway that does not exist today. Pilot timelines cut from 3–5 years to 12–18 months. Pre-negotiated permitting, finance, and community frameworks available from day one. Learning opportunity for Japanese founders to engage with world-class technology; potential spin-offs or feeder technologies
Foreign innovators	Japan as a deployment-ready priority market — pre-built utility access, permitting pathways, and JOGMEC financing in place before the first meeting. Currently, complexity alone deters entry.
Corporates (utilities, steel, chemicals)	Structured interface to engage pre-commercial climate technology aligned with 2030 GX targets. PPA-structured co-investment limits balance-sheet exposure at pilot stage.
Government / METI / Prefectures	Policy intent translated into operational pilot delivery. Replicable community engagement methodology replacing improvised, frequently unsuccessful ad-hoc approaches.

FIVE SECTORS, ONE OPERATING SYSTEM

Sector	OS Role	Transfer Logic
01 Geothermal	Anchor — OS v1	All four modules built here. Every subsequent sector inherits the Permit Router, Community Playbook, and Finance Stack.
02 Methane Monitoring	Trust-builder	Strong JAXA/NEC satellite capability; LNG corporates under Scope 1 pressure. Ideal low-barrier entry before harder pilots.
03 Industrial Heat	OS inheritance	Steel, chemicals, ceramics. Permit Router inherits hydrogen-adjacent precedents; Finance Stack adapts corporate PPA structures.
04 Long-Duration Storage	Grid adaptation	Island grid creates acute multi-day storage demand. Permit Router adapts the grid-connection decision tree; Finance Stack uses the utility offtake structure.
05 Carbon Capture	Playbook transfer	~30% of Japan's CO ₂ in cement, steel, chemicals. Community Playbook transfers directly — geological storage opposition mirrors onsen-operator resistance.

And many more to build

THREE-YEAR PROGRAMME BUDGET: \$10–15M

The philanthropic ask catalyses ~3×+ from NEDO, JOGMEC, and corporate PPA partners. OS infrastructure — the component that compounds in value with every new pilot — is under one-third of total spend.

%	Category	Amount	What It Funds
33%	Pilot Co-funding	~\$3–5M	4–6 pilot grants (\$500K–1M each) structured to unlock 3× NEDO/corporate co-investment per dollar across hydrothermal, AGS, and supercritical tracks
27%	Deployment OS Build	~\$2–4M	Permit Router, Community Playbook, and Finance Stack v1 — built once; marginal cost per pilot falls sharply from Year 2
25%	Core Team	~\$2–4M	5–7 FTEs: Japan geothermal regulatory specialists, onsen-community engagement leads, programme managers, and an ex-METI/JOGMEC senior advisor
15%	Learning Loop + Comms	\$1–2M	Structured knowledge capture from every pilot, public case studies, and METI/NEDO/JOGMEC relationship infrastructure

DEPLOYMENT GATES

Period	Hard Success Criteria
Year 1 (\$3–4M) Build & Seed	~6 FTEs in place including ex-METI/JOGMEC senior advisor and kumiai liaison · Permit Router v1 across 5+ prefectures · Playbook piloted with 2 Onsen Kumiai · 3 pilot MOUs signed · NEDO/JOGMEC co-financing agreed for Pilot 1
Year 2 (\$4–5M) Deploy & Test	3–5 active pilots across hydrothermal, AGS, supercritical · First formal kumiai consent letter under Playbook v1 · Sub-12-month permit demonstrated · Corporate PPA with named offtaker · Learning Loop v1 live
Year 3 (\$3–4M) Scale & Prove	First generating well online · Supercritical well ≥10 MW output · OS cost ≥40% below Y1 baseline · OS v2 adopted by METI/JOGMEC · Replication roadmap for second sector published

KEY RISKS & MITIGATIONS

Level	Risk	Description	Mitigation
HIGH	Adoption	No OS without pilots; no pilots without OS. Securing 2–3 anchor partners before the philanthropic runway expires is the single most critical near-term task.	Active Y1 MOU pipeline in development. Methane monitoring used as a lower-barrier entry point to build corporate relationships ahead of harder pilots.
MED	Regulatory grey zone	CLGS has no dedicated legal category under the Hot Springs Act. All 47 prefectures currently interpret the rules independently.	METI engagement from Year 0. The Next-Gen Geothermal Council (70+ companies) is the primary forum for clarification and pre-competitive alignment.
MED	Team capacity	Deep knowledge of Japan's regulatory environment is rare. Managing 3–5 simultaneous pilots requires strong coordination infrastructure.	ARCS concentrates this knowledge rather than leaving it fragmented. Learning Loop reduces per-pilot burden from Y2. Ex-METI/JOGMEC advisor in the core team.

Above is a draft proposal on enabling the deployment of ARCS – actively encouraging feedback and discussions to improve the set-up.